

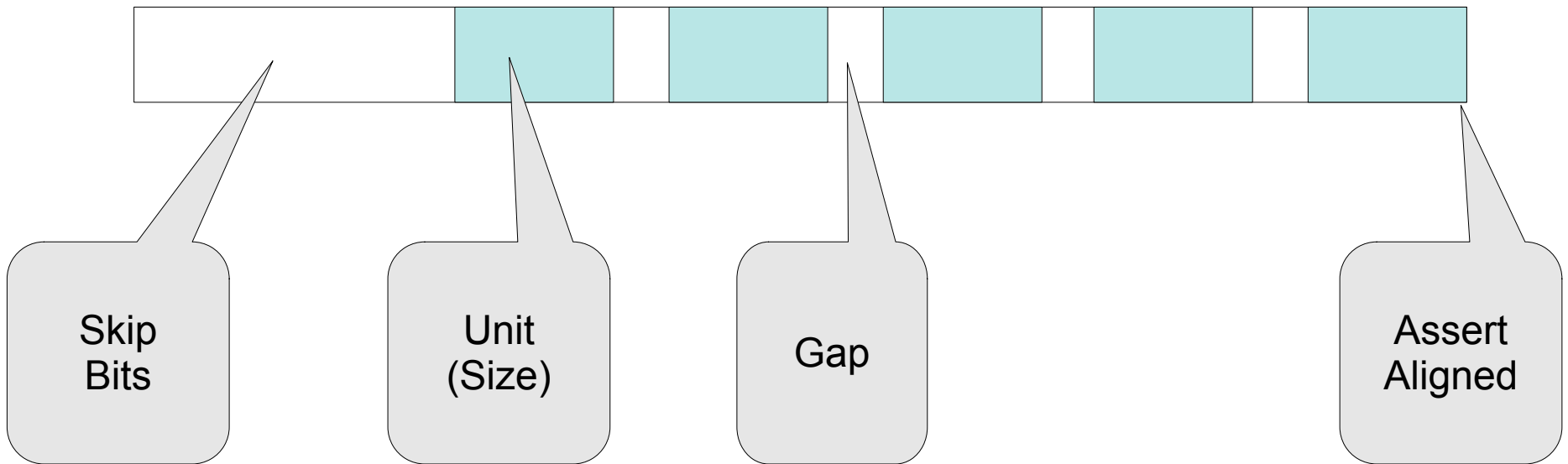
# bdd

(<http://bdd.sourceforge.net>)

Bit Stream Manipulation Command  
For Unix/Linux/POSIX Command Line  
Implemented in Python  
Very Flexible – Not Very Efficient  
Freeware

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# A Bit Stream

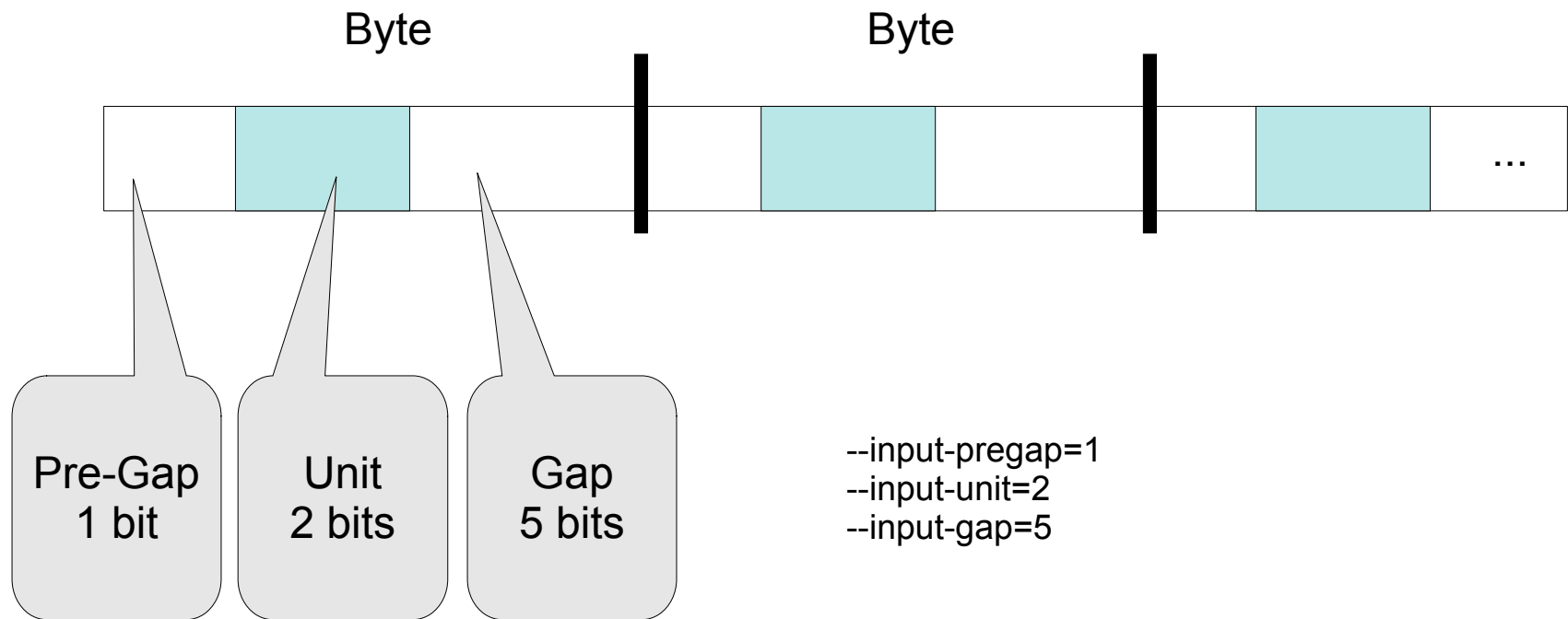


All measures in bits, not bytes

e.g.  
`--input-skip-bits=128`  
`--input-unit=7`  
`--input-gap=3`  
`--input-assert-aligned`

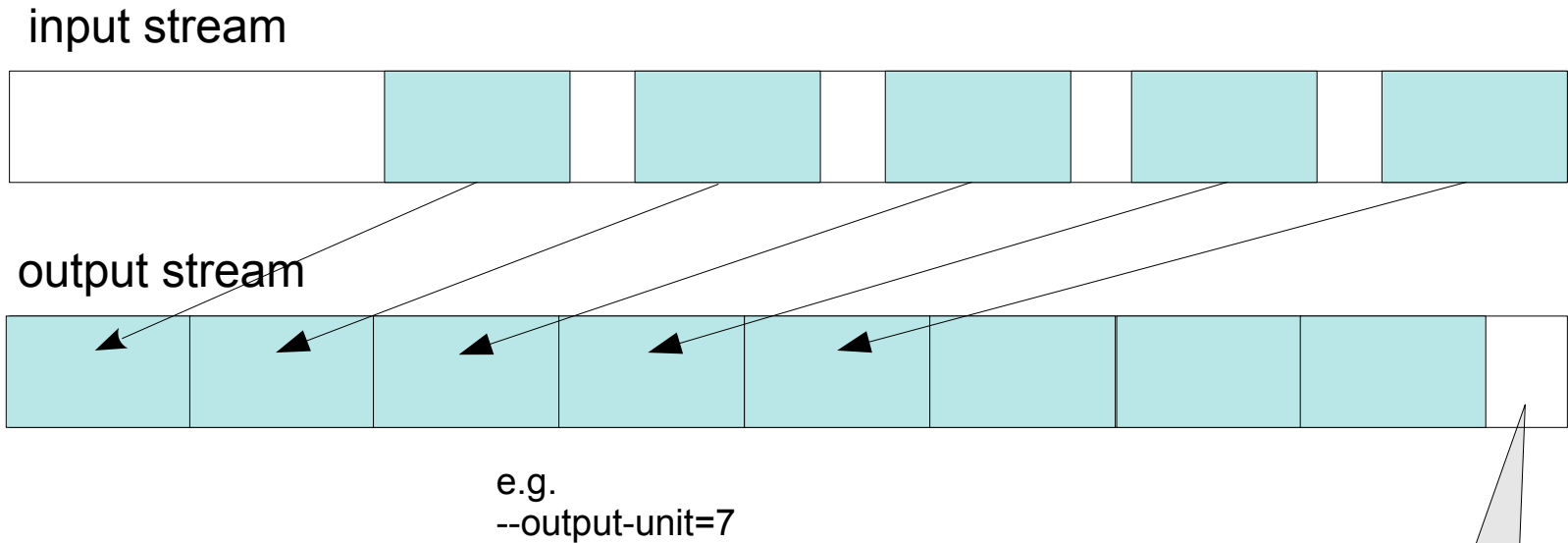
Gives an error message if not aligned to a byte

# Example, Bits from a Byte Stream



--input-pregap is more convenient but identical to:  
--input-skip-bits <pregap more>  
--input-gap <pregap more>

# Output Bit Stream



If the size of the output unit is bigger  
than the input unit:

Left bits are truncated

If the size of the output unit is smaller  
than the input unit:

Left bits are filled with zeros

Unit is handled as if  
it was an integer

Last Byte  
Filled with  
Zeros

# Example, Rearrange Bit Streams

- Take most significant bit of a byte stream
  - `bdd --input-unit=1 --input-gap=7 \`  
`--output-unit=8`
    - examples assume that the input is piped to the bdd like “cat file | bdd”, you can also use `--input-file`
    - 1 bit will be filled to 8 bits: most significant bits = 0
    - Output byte stream of values 0x00 and 0x01
- Pack least significant bit of bytes to a bit stream
  - `bdd --input-pregap=7 --input-unit=1 \`  
`--output-unit=1`
    - If input is 9 bytes all bits ones, the output will be two bytes 0xFF and 0x80

# “Fake” Bit Streams

- Input:
  - Zeros
  - Ones
  - Random
    - /dev/urandom
  - Counter
    - Starts from zero
    - Every unit previous plus one
- Output:
  - Print unit as integer
    - one integer at line
    - meant for programs
  - Print unit as hex
    - human-readable lines
  - Print unit as bits
    - human-readable lines

# Default Unit Size

- Rule to remember:

`bdd <input >output`

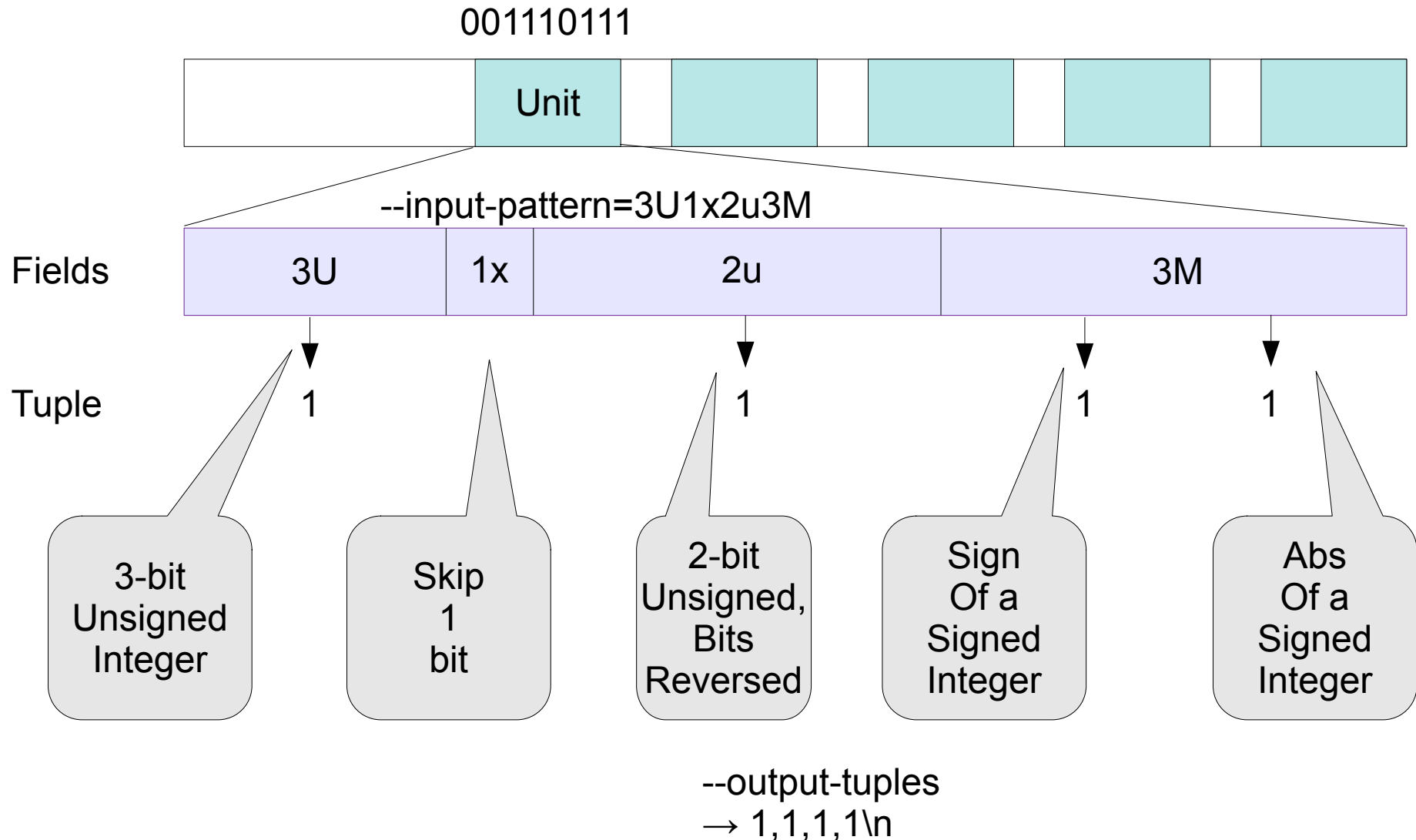
just copies 8-bit bytes from input to output

- Default input unit is 8 bits
- Default output unit is 8 bits  
(will likely change in the future:  
the same as the input unit)

# Tuple

- Tuple is a unit that is split into fields
- It is possible to give fields types so that e.g. some bits are interpreted as signed integers or floating point numbers

# Tuple Unpack



# Pattern Characters (Input)

- U – Unsigned integer
  - x – Skip bits
  - M – Signed integer as two unsigned integers
- 
- U can be used to unpack any bit pattern and it will pack identically
  - M assumes that the field is 2's complement integer
  - M will return two values: first the sign (0=positive, 1=negative) and then abs(value)

# “Fake” Tuple Streams

- `--input-tuples` reads human-readable tuples directly from the input stream
- `--output-tuples` writes human-readable tuples directly to the output stream
- Comma separated fields, one line per tuple:
  - `1,1.2,"abba"`

# Example, Look inside a Bit Stream

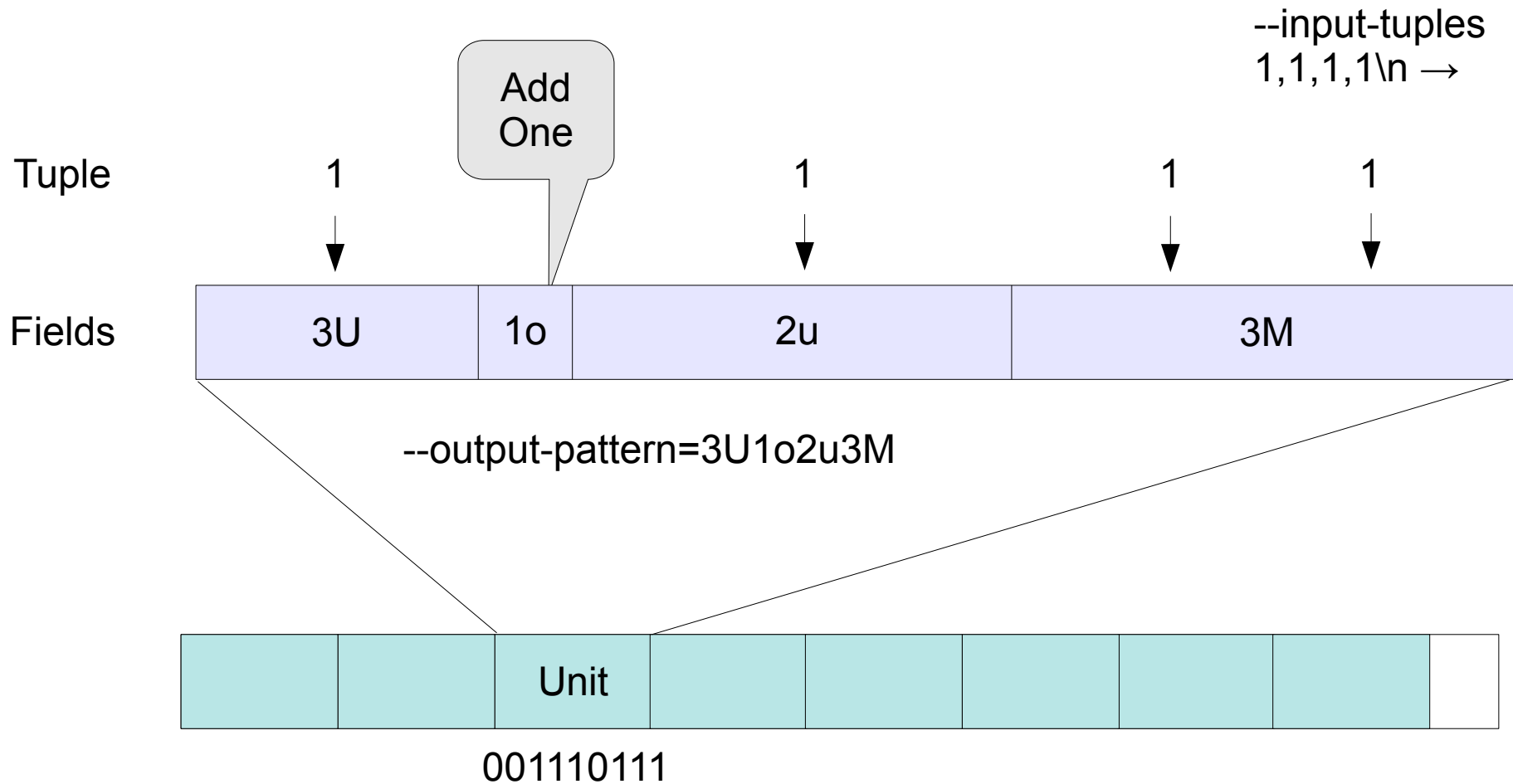
- Interpret units to human-readable lines:
  - `bdd --input-pattern=3U1x2u3M --output-tuples`
  - Prints result as comma separated list, one unit at a line, e.g.

```
1,1,1,1  
3,2,0,2  
...
```

# Typed Pattern Characters

- The following pattern characters do not return integer value. They print nicely with --output-tuples, but if you try to repack or manipulate them, pattern must exactly match the type.
- S – signed integer
- F – 32-bit IEEE floating point number
- D – 64-bit IEEE floating point number
- C – characters (printed as UTF-8)

# Tuple Pack



# Pattern Characters (Output)

- Output patterns have some special characters:
- z – Fill bits with zeros
- o – Fill bits with ones
- r – Fill bits with random bits (/dev/urandom)

# Examples, Pack from Text

- Build a hex number from integers:

- `echo -en "1,2\n3,4" | bdd --input-tuples \`  
`--output-pattern=4U4U | od -t x1`
    - Prints (hex) 12 34

- Build a stream of floating point numbers:

- `echo -en "1.0\n2.0\n" | \`  
`bdd --input-tuples --output-pattern=64D`

# Example, Combined

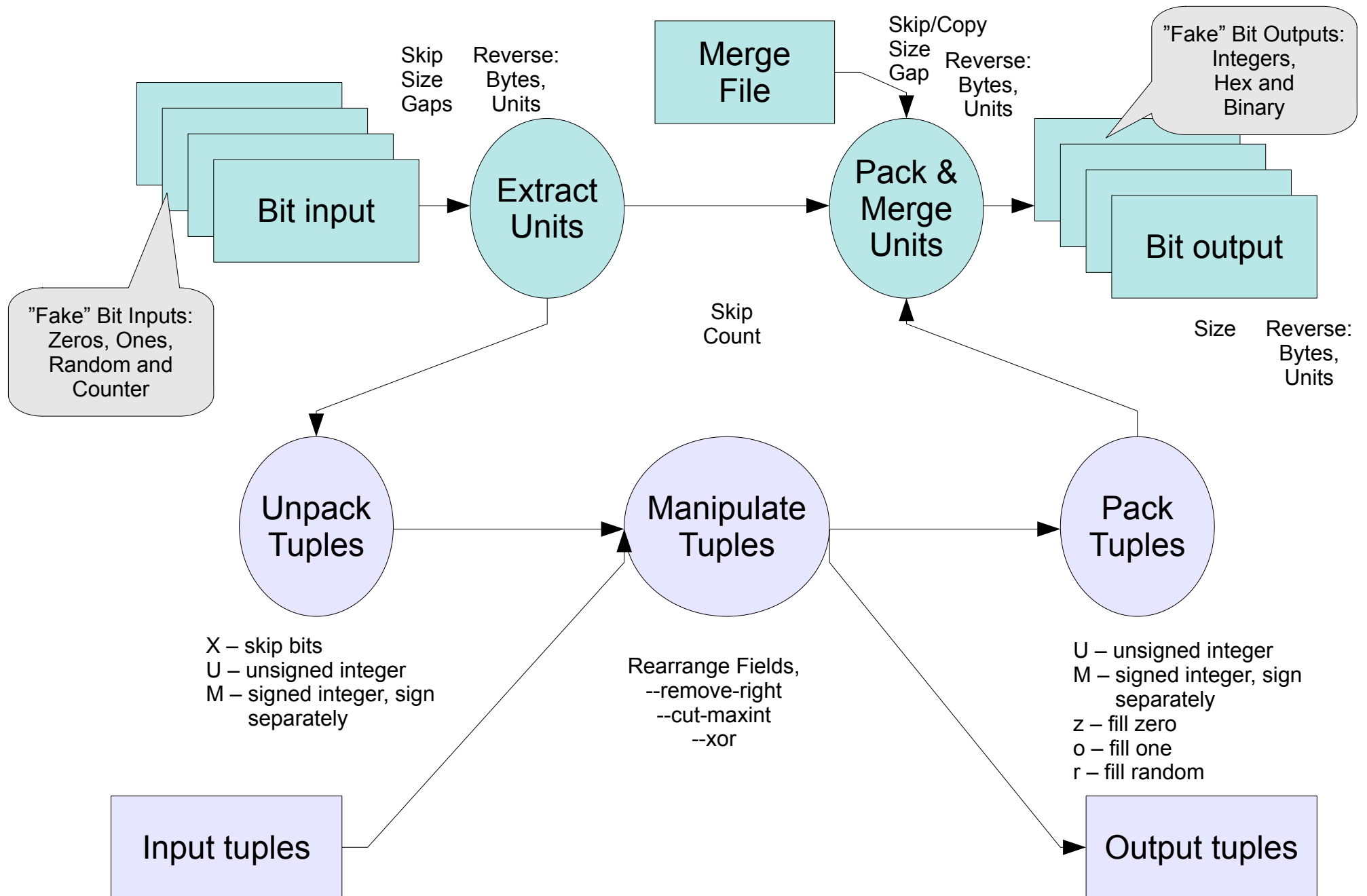
```
→ echo 1,1,1,1 | \  
  bdd --input-tuples --output-pattern=3U1o2u3M | \  
  bdd --input-pattern=3U1x2u3M --output-tuples
```

- **Prints:**

- 1,1,1,1

- 0,0,0,0

- 9-bit output pattern outputs two bytes and when read, the second is read as all-zeros



# Example, Manipulating Tuples

- Input of 1+2 bits in stream, turn bits 2+1
  - `bdd --input-pattern=1U2U \`  
`--rearrange=1,0 \`  
`--output-pattern=2U1U`
    - Fields are numbered starting from 0
- Input is signed bytes, we want the most significant bit of the magnitude, after -128 is truncated to -127:
  - `bdd --input-pattern=8M \`  
`--rearrange=-1 --cut-maxint=0,127 \`  
`--remove-right=0,6 --output-unit=1`
    - -1 is the last field in tuple, -2 second last etc.
- Options are executed in the specified order

# Reversing Bits

- Reversing bits can be quite a mess...
- bdd implements
  - Reversing bits of every byte read
  - Reversing bits of every unit read
  - Matching patterns in reverse bit order
  - Reversing bits of every unit written
  - Reversing bits of every byte written
- --in/output-little-endian is a synonym for combined reversing of both bytes and unit

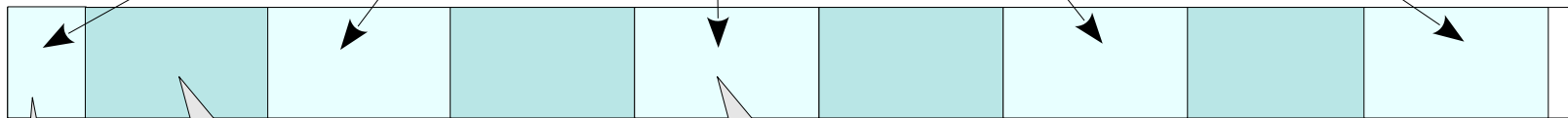
# Merge Bit Stream

Merge file options  
are mostly identical  
to input file options:  
skip, gap, reverse

Merge stream



Output stream



Normal  
Unit

Merge  
Unit

“Copy  
First”

A merge unit is written after each normal unit  
It is an error if merge file ends too early  
Output will have exactly as many normal and merge units

# Example, Merge

- Print first 16 bytes of /etc/passwd file as hex preceded by a three-digit hex address of the byte:

```
→ bdd --input-counter --count=16 \  
    --input-unit=12 --output-unit=12 \  
    --merge-file=/etc/passwd \  
    --output-hex
```

```
000 72 001 6f 002 6f 003 74 004 3a 005 78 006 3a 007 30  
008 3a 009 30 00a 3a 00b 72 00c 6f 00d 6f 00e 74 00f 3a
```